

The connectivity dilemma with the expansion of IIoT has exposed security gaps. This is depicted in Figure 1.

Many OT systems are externally accessible, with public-facing applications and internet accessibility, making them susceptible to attacks. Vulnerabilities in industrial control systems (ICS) have increased by 50%, with 61% being remotely exploitable. The loss of availability, a critical element in the CIA triad (confidentiality, integrity, and availability), is a major concern, with 65% of attacks leading to downtime. Outdated operating systems like Windows XP and Windows 7 pose additional risks. Many organisations still use these unsupported systems, making them easy targets for exploitation. Furthermore, ransomware attacks targeting specific industries have become a significant menace.

As we continue to embrace the potential of Industrial IoT, it is crucial to address the security challenges that come with it. The convergence of OT and IT systems demands a comprehensive approach to cybersecurity. Organisations must prioritise securing their IIoT infrastructure to protect against potential threats and ensure the uninterrupted operation of critical systems. In this era of connectivity, security is not an option but a necessity. With vigilance, education, and robust cybersecurity measures, we can safeguard our Industrial IoT assets and pave the way for a more secure and prosperous future. As organisations worldwide embrace the opportunities presented by digital transformation, they must also be acutely aware of the risks that come hand in hand.

The changing face of threats

Digital transformation has ushered in a new era of threats. Ransomware attacks have escalated in both frequency and intensity, posing a significant danger to businesses. Even insider threats have

Notable industrial cyberattacks and the growing threat landscape in IIoT

In the world of Industrial IoT (IIoT), the potential for catastrophic cyberattacks cannot be underestimated. Here are some examples that highlight the devastating consequences of such attacks:

- **The Ukraine power grid attack:** Initiated by a spear-phishing email targeting a high-profile individual, this attack compromised both IT and OT systems, culminating in a network takeover. In just six hours, 230,000 people faced a devastating power grid shutdown amid freezing temperatures.
- **Saudi oil refinery attacks:** Targeted critical infrastructure in oil and gas, nuclear, and manufacturing sectors, posing a grave threat to national security and international stability.
- **Norsk hydro ransomware attacks:** Causing disruptions in aluminium production across approximately 170 plants, these attacks led to significant financial losses, highlighting the vulnerability of critical industrial infrastructure to cyber threats.
- **Colonial pipeline hack:** Resulting in widespread fuel and gasoline shortages, this cyberattack disrupted the crucial energy supply chain, leading to fuel shortages at gas stations, and underscoring the vulnerability of critical infrastructure to cyber threats, thereby raising concerns about national security and resilience.

However, new attacks are emerging every day. A 2021 report indicated 10 successful ransomware attacks on the manufacturing segment. The potential loss is staggering, considering the encryption of data and shutdown of machines. As the IIoT landscape expands, so does the threat landscape. The lessons learned should serve as a clarion call for heightened vigilance, rigorous training, and stringent security protocols to protect against future cyber onslaughts.

become a cause for concern, as enemy states can manipulate individuals within an organisation to compromise information or alter configurations, leading to severe consequences. The expanding attack surface is a critical concern. The convergence of IT (information technology) and OT (operational technology), the widespread adoption of 5G and LTE, and the migration to cloud computing have all increased the potential vulnerabilities. Organisations that do not configure their cloud resources securely expose themselves to a host of security risks.

By 2032, an astonishing 2.6 billion OT assets will be connected to the internet. However, a startling fact is that more than 47% of organisations lack the necessary security tools to effectively counter these threats. This glaring security gap underscores the urgent need for cybersecurity measures. Furthermore, a staggering 94% of industries have experienced a security incident in the past 12 months. While this statistic may not be surprising given the ever-increasing threat landscape, it serves as a stark reminder of the constant and growing danger that the organisations face.

Common attack vectors

Several attack vectors persist despite the advances in cybersecurity. Web application attacks, external hardware or removable media (such as USB drives), and compromised remote access are among the prevalent threats. Data theft and ransom-based attacks continue to plague organisations. Moreover, the proliferation of APIs (application programming interfaces) has opened new avenues for cybercriminals to exploit. In the realm of Industrial IoT, threat actors come in various forms. Negligent insiders, whether intentionally or unintentionally, remain a significant concern. Phishing attacks targeting employees, including top executives, are rampant. Cybercriminals driven by profit motives, rival companies seeking an edge, and malicious insiders coerced into compromising an organisation's security all play a role in the evolving threat landscape. Additionally, hacktivists, motivated by specific agendas, can wreak havoc, and nation-state actors pose a grave danger, especially in the context of geopolitical conflicts.

The industrial control system Kill Chain offers insight into the stages of a cyberattack in this domain. It starts with reconnaissance, where hackers gather information. Next comes weaponisation,